Analytical Cume--November 14, 1992

Several weeks ago, Dr. Charles A. Evans of Charles Evans and Associates (Redwood City, CA) presented to us an Analytical Chemistry seminar (in the Bell Engineering building) on a surface characterization method called SIMS.

1. SIMS, general questions:

(8 pts.) (a) What do the letters "SIMS" stand for?

(8 pts.) (b) What surface information does SIMS reveal?

(16 pts.) (c) In general, how does SIMS work? Draw a diagram of a surface to help in your explanation and indicate primary and secondary ions. Show the molecular fragmentation and molecular desorption regions of the sample.

2. The SIMS that Dr. Evans described has a Time-of-Flight (TOF) mass analyzer (as opposed to a magnetic sector or quadrupole mass analyzers).

(11 pts.) (a) Briefly describe how TOF separates secondary ions of different masses.

(**Hint**: all the secondary ions obtained with a given pulse of primary ions have essentially the same kinetic energy, **E**, where

$$E = eV = (1/2)\ M\ v^2,$$

M = mass in kg/mol, **v** = velocity in m/s, **e** = charge = **$1.6021892 \times 10^{-19}$ C**, **V** = electric potential in volts.)

(11 pts.) (b) Given a molecular fragment of $CH_3CH_2^+$ (**$M = 29 \times 10^{-3}$ kg/mol**) that is accelerated through a **10 V** electric potential, what is the transit time of the fragment across a TOF tube that is **1 m** in length?

3. Discuss the complications or difficulties encountered with SIMS in the analysis of the following materials. How do these complications affect the mass spectra that are obtained from these materials?

(5 pts.) (a) insulating materials

(5 pts.) (b) polymeric materials

4. SIMS is carried out in a high vacuum chamber. There are several reasons for this.

One reason for the high vacuum is that at low pressure, there are fewer collisions of the residual gases with surfaces, and thus, less contamination of the surfaces with time. This is extremely important for studies where surface cleanliness is important.

The pressure, **P**, may be related to **z, the number of collisions the molecules make on a surface in terms of per unit area per unit time (the units of z are $m^{-2}\ s^{-1}$)**. The following equation demonstrates this relationship

$$P = \frac{z\,(2\ \pi\ M\ R\ T)^{1/2}}{N_A}$$

where **M** is the mass of the molecules in kg/mol, **R** is the ideal gas constant in $JK^{-1}mol^{-1}$, **P** is the pressure in Pascals (1 Pa = 1 N/m^2), **T** is the temperature in Kelvin, and $\mathbf{N_A}$ is Avagadro's number.

The lower the pressure, the fewer the collisions in the vacuum chamber, and the greater the mean free path. (In other words, the average distance that a molecule or ion travels between collisions is large.) The larger the mean free path, the less the contamination of the sample surface with residual gases. Below is the equation of the **mean free path** (in meters).

$$\lambda = \frac{R\,T}{\sqrt{2}\,\pi\,d^2\,N_A\,P}$$

where **d** is the collision diameter of the molecule in meters.

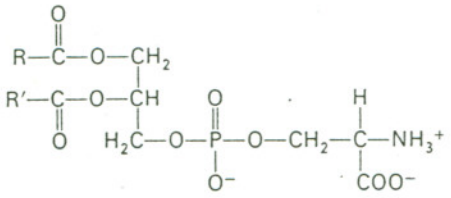
(10 pts.) **(a)** What pressure (in torr) of molecular oxygen is required to achieve a **0.1 monolayer/sec** contamination rate of a surface at **300 K**? (The pressure that you will calculate is typical for ultra high vacuum instruments.)

Information that will be useful:

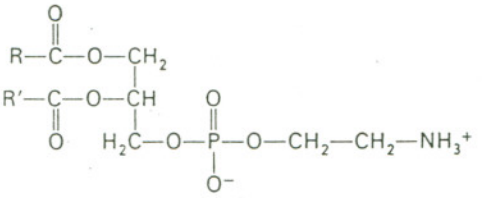
mass, **M**, of molecular oxygen = **32×10^{-3} kg/mol**
R = 8.314 J K^{-1} mol^{-1}
d (molecular oxygen) = **3.57×10^{-10} m**
Assume that **1 monolayer = 1×10^{-9} mol/cm^2**.
Assume that *every* collision event results in contamination of a single site on the surface.
A single surface site is equal to the area occupied by a single molecule.
1 Pa = 1 N/m^2
1 N = 1 kg m/s^2
1 torr = 133.322 N/m^2
1 mol = 6.022×10^{23} molecules

(10 pts.) **(b)** What is the mean free path of the molecules at the pressure that you calculated in part (a)?

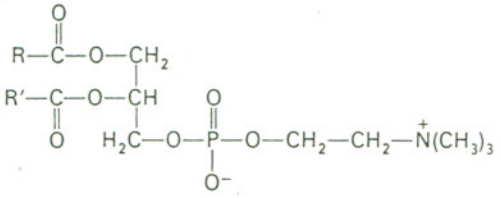
(16 pts.) **(c)** Suggest two *more* reasons (in *addition* to the one described above) why high vacuum is necessary in SIMS. (Think about how SIMS works, what it analyzes, and what problems SIMS would have if operated at atmospheric pressure).



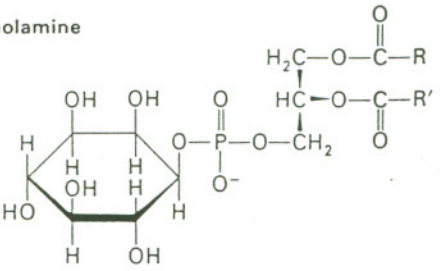
Phosphatidyl serine



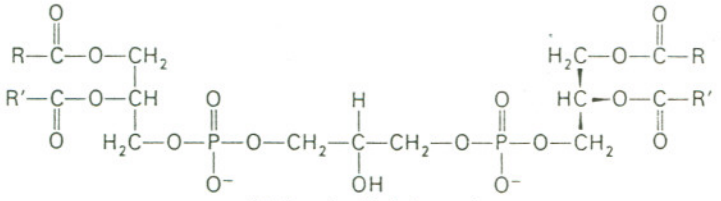
Phosphatidyl ethanolamine



Phosphatidyl choline



Phosphatidyl inositol



Diphosphatidyl glycerol
(Cardiolipin)

Figure 12-5
Formulas of some phosphoglycerides.